

NOAH PEARL

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EDUCATION

University of California, Santa Barbara

Expected June 2027

B.S. in Statistics and Data Science · GPA 3.75 · Dean's List

Probability & statistics: Probability and Statistics I-III (PSTAT 120A/B/C), Regression Analysis (126), Generalized Linear Models and Nonparametric Regression (127), Statistical Machine Learning (131), Experimental Design (122), Stochastic Processes (160A/B), Data Science Principles (100)

Mathematics: Calculus sequence, Linear Algebra, Transition to Higher Mathematics (PSTAT 8)

In progress, Fall 2026: Methods of Analysis (MATH 117), Survival Analysis (PSTAT 175), Database Management (PSTAT 132)

RESEARCH EXPERIENCE

ISI-BUDS, University of California, Irvine

June - August 2026

Biostatistics and Undergraduate Data Science Summer Institute - NIH-funded research experience for undergraduates

Mentors: Prof. Volodymyr Minin (Statistics), Prof. Daniel Parker (Public Health), Thanasi Bakis (PhD candidate)

Project: Environmental predictors of *Aedes aegypti* abundance in the Coachella Valley, in partnership with the Coachella Valley Mosquito and Vector Control District.

- Built the analysis dataset joining ~17,900 BG-Sentinel trap collections across 1,322 locations (2016-2025) to satellite and reanalysis covariates (vegetation and surface water indices, land cover, precipitation, temperature) summarized over 200 m buffers and rolling 14-day windows, in R with Google Earth Engine products.
- Characterized spatial and seasonal structure across a decade of surveillance data, including a September-October peak and year-over-year growth in abundance independent of sampling effort, with effort modeled as a log offset for trap-nights.
- Built a gradient-boosted count model as a benchmark for the team's negative binomial GAM under a matched covariate design and leave-one-year-out cross-validation over 2018-2025. The two model classes produced nearly identical error curves and systematically under-predicted peak years, evidence that the missing information was in our covariates.

PRESENTATIONS AND REPORTS

Hirko, L., Yao, L., and Pearl, N. (2026). Environmental predictors of *Aedes aegypti* abundance in the Coachella Valley. ISI-BUDS Summer Symposium, University of California, Irvine, August 2026.

Pearl, N., Hirko, L., and Yao, L. (2026). Environmental predictors of *Aedes aegypti* abundance in the Coachella Valley. Unpublished technical report, ISI-BUDS, University of California, Irvine.

PROJECTS

Predicting NBA Regular-Season Win Totals

June 2026

- Compared four model families predicting NBA win totals (805 team-seasons) and showed box-score fundamentals alone match models that use near-tautological margin metrics.

LEADERSHIP AND EMPLOYMENT

UCSB AI / Machine Learning Club - Co-founder and VP of Projects

Sep 2024 – March 2025

- Guided group project on retrieval-augmented generation, Hugging Face model use, and AI safety and ethics.
- Grew the club past 100 members in its first two quarters.

The Ritz-Carlton Bacara, Santa Barbara - Server

May 2025 – April 2026

Sigma Pi Fraternity - Philanthropy Chair

Jan 2024 – June 2025

TECHNICAL SKILLS

Statistical computing: R (tidyverse, tidymodels, mgcv, glmnet, ranger, xgboost, terra, LightGBM, lubridate, Quarto), Python (pandas, NumPy, seaborn, statsmodels, scikit-learn), SQL

Tools: Git/GitHub, Google Earth Engine, Java